



Khulna University of Engineering & Technology

Department of Computer Science and Engineering

Speaker-Attributed Financial Discourse Analysis of Bangla Talk Shows

A Five Stage Diarization, Recognition and Target-Aware Sentiment Pipeline

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Course Title:

Machine Learning Laboratory

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Corpus:

26 episodes, 16.88 hours

Submitted By

Roll: 2107001

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1. Abstract

This report documents a five stage pipeline that turns raw Bangla business talk show audio into speaker-attributed polarity judgements about named financial targets. The speech path conditions broadcast audio to 16 kHz mono at a controlled loudness, segments it by speaker with pyannote, transcribes it with a Bengali adapted Whisper model in 28 second chunks, and fuses words to speakers by timestamp overlap, producing 3,532 speaker-attributed utterances. The language path detects financial targets with a 52 entry stem lexicon, assigns polarity with a target conditioned rule set, and trains a classifier on the resulting labels.

Evaluation uses 180 target pairs read and labelled by hand. The rule set reaches a macro F1 of 0.584 and a Cohen kappa of 0.392 against those labels, while a character n-gram classifier trained on its output reaches 0.466 on held out episodes, below the rule set that supervised it. The report also records two transcript defects found by direct inspection, affecting 11.3 and 1.3 percent of utterances, and identifies the label rules rather than transcript noise as the current bottleneck.

2. System Architecture

The system is a cascade of five stages. Each stage reads the durable output of the previous one and writes its own, so any stage can be rerun, inspected or replaced without disturbing the others. No state is held in memory between stages.

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Audio	Diarization	Recognition	Fusion	Polarity
16 kHz mono	1,867 turns	130,750 words	3,532 utterances	3,109 pairs

Fig 1: The five stage cascade

Stages 2 and 3 are independent of each other: diarization never sees the transcript and recognition never sees the speaker labels. This is a deliberate design property. A failure in one does not corrupt the other, and either can be rerun in isolation. It also means the two stages disagree about time boundaries, and reconciling that disagreement is the entire job

of stage 4, which is the first point where both are required and therefore the point where errors compound.

3. Data

The corpus is 26 full episodes of Bangla business and economics talk shows drawn from six broadcasters, contributed by six group members. Episodes run 24 to 48 minutes. The material is deliberately difficult: panel discussion with frequent crosstalk, studio music between segments, guests joining by telephone at reduced bandwidth, and heavy code switching in which English finance terms appear inside Bangla sentences.

Roll	Episodes	Words	Programmes
2107001	4	21,500	Business Talk, ATN
2107004	6	29,383	Star Biz, Dhaka Stream
2107006	5	25,736	RTV Business Talk
2107009	5	23,447	Money Talks, Desh TV
2107010	1	2,960	Star Biz Dialogue
2107015	5	27,724	ATN, Gtv
Total	26	130,750	16.88 hours

Table 1: Corpus composition by contributor

Episode identifiers are namespaced by roll number, so 2107001::ep001 and 2107006::ep001 remain distinct entities throughout. Speaker identifiers are episode local by construction and are never merged across recordings: speaker 2 in one episode has no relationship to speaker 2 in another. They are cluster indices, not people.

4. Preprocessing: Audio Conditioning

Both downstream speech models expect 16 kHz mono at a predictable loudness. Broadcast MP3 satisfies neither condition, and levels vary substantially between channels. Stage 1 resamples 44.1 and 48 kHz stereo sources to 16 kHz mono, then measures integrated

loudness to the EBU R128 standard against a target of -16 LUFS. Measured values across the corpus span -28.3 to -16.0 LUFS.

Gain is applied selectively rather than universally. An episode is adjusted only when it is genuinely off target and its true peak leaves headroom; roughly one third of measured episodes were changed and the rest were left untouched. Normalising every file blindly would clip the loud ones and introduce distortion that the speech models would then have to work around.

Why loudness matters two stages later

Diarization clusters speaker embeddings by distance. If one episode arrives 12 dB quieter than another, its embeddings shift and a clustering threshold tuned on the first recording stops working on the second. Controlling loudness first is what allows one threshold configuration to apply across all 26 episodes rather than requiring per episode tuning.

5. Speaker Diarization and Postprocessing

Stage 2 uses the pyannote speaker-diarization-community-1 pipeline with the default speaker embedding replaced by wespeaker-voxceleb-resnet34-LM, which separates voices more reliably on noisy broadcast audio. Voice activity uses a minimum off duration of 0.1 s and clustering uses a minimum cluster size of 20.

The raw output of any diarizer on broadcast audio is fragmented and contains spurious speakers, so a postprocessing pass is applied. Both the raw and the postprocessed results are retained, so the effect of each threshold can be inspected rather than assumed.

Postprocessing step	Threshold	Purpose
Minimum segment duration	0.75 s	A cough or a jingle does not become a turn
Gap stitching	0.17 s	One sentence does not split into three turns
Segment merge threshold	3.79	Adjacent turns of one speaker are combined
Minimum speaker airtime	9.0 s	Spurious clusters are deleted entirely
Scoring collar	0.25 s	Standard allowance for boundary imprecision

Table 2: Diarization postprocessing thresholds

The stage produces 1,867 speaker turns across the corpus, with a median of 3 speakers per episode and a range of 1 to 7, which matches the expected format of a host with two or three panellists.

No diarization error rate is reported

Diarization error rate requires a human made reference of who spoke when. No such reference exists for this corpus, and scoring automatic output against itself would be meaningless. The figure is left blank rather than estimated. Section 12 gives the coverage statistic reported in its place and states precisely what that statistic does and does not claim.

6. Speech Recognition

Stage 3 uses a Bengali adapted Whisper medium checkpoint converted to CTranslate2 and run in float16 on GPU. Whisper reads a fixed 30 second receptive field, so transcribing a 40 minute episode is fundamentally a chunking and stitching problem wrapped around a short context model. The corpus was decoded in 2,184 chunks producing 4,006 segments and 130,750 words with word level timestamps.

Setting	Value	Reason
Chunk length	28.0 s	Leaves headroom inside the 30 s field, so the window edge never sits at the model limit
Language	bn, forced	Stops the detector switching to Hindi or Assamese on a noisy chunk
Beam size	5	Standard accuracy and speed compromise
Condition on previous text	False	Stops a hallucination in one chunk entering the next as context
Source separation	On detection	Spectral flux is measured per chunk; music is separated only when present
Repetition penalty	0.8	Below 1.0, which encourages repetition rather than suppressing it. See section 7.2

Table 3: Decoding configuration as recorded in the run manifest

Disabling conditioning on previous text is the most consequential of these choices. With it enabled, a single hallucinated phrase is fed forward as context and the decoder can produce a corrupted minute of transcript from one bad chunk. Disabling it costs some fluency across boundaries but bounds the damage of any single failure to the chunk that produced it.

The choice of a Bengali adapted rather than stock multilingual checkpoint matters most for the financial vocabulary the rest of the system depends on. It produces মূল্যস্ফীতি, খেলাপি ঋণ and বৈদেশিক মুদ্রার রিজার্ভ as clean tokens rather than phonetic approximations, which the target lexicon in section 9 requires in order to match anything at all.

7. Transcript Quality Audit

Every stage reported success, so the transcripts were inspected directly rather than trusted. Two systematic defects were found and quantified across all 3,532 utterances.

7.1 Chunk boundary truncation

Bangla orthography never begins a word with a dependent vowel sign. When a segment does, the chunk stitcher has removed the opening consonant. This makes detection mechanical rather than a matter of judgement, so the count is exact: 400 utterances, or 11.3 percent of the corpus, begin with a token that is not a valid word.

ুক্তা উন্নতি লক্ষ্য করা যাচ্ছে

The opening word should be কিছুটা. The leading consonant has been consumed at a chunk boundary. Worst affected episode: 21.2 percent of its utterances.

The cause is that chunks are cut back to back with no overlap. Cutting with a 2 to 3 second overlap and stitching on the shared region removes the boundary word entirely. This is a rerun configuration change, not new code.

7.2 Decoder repetition loops

In 46 utterances, or 1.3 percent, the decoder locks onto a single token and emits it for many seconds. Detection uses two signals together: the longest run of one repeated token, and the unique token ratio across a long span. The run catches a single repeated token and the ratio catches an alternating loop.

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These carry no readable meaning. They are excluded from supervision rather than quietly labelled neutral, which would teach the model that meaningless text is non evaluative.

Probable cause

The run sets a repetition penalty of 0.8. A penalty above 1.0 suppresses tokens that have already been produced; a value below 1.0 does the opposite and makes repeating them more likely. Raising the value above 1.0, or adding a no repeat n-gram constraint, is the direct remedy. This has not yet been confirmed by a rerun, so it is reported as the probable rather than the established cause.

8. Fusion: Attaching Words to Speakers

Diarization produces time intervals with speaker labels. Recognition produces words with timestamps. Neither knows about the other, so stage 4 must decide which speaker owns each word.

1. Take the midpoint of the word. If it falls inside one speaker interval, that speaker owns the word.
2. If the midpoint falls in a gap between intervals, fall back to the interval with the largest temporal overlap with the word.
3. If two speaker intervals overlap in time, the speaker who started first keeps the word, so no word is ever counted twice.
4. Consecutive words from the same speaker are grouped into an utterance, which becomes the unit of analysis for everything downstream.

The midpoint is used rather than the word start because start timestamps drift at chunk boundaries, where the truncation defect of section 7.1 also originates, and the midpoint is the more stable of the two anchors. The result is 3,532 speaker-attributed utterances carrying speaker, timing and text together, which is what allows any later polarity judgement to be traced back to a specific person at a specific second of a specific episode.

Word to speaker attribution reaches 99.5 percent across the rolls with diarization completed. This is a coverage statistic: it states that a word was attached to some automatic speaker interval. It is not a claim that the attributed speaker is correct, which would require the human reference discussed in section 5.

9. Target Detection

Polarity is meaningless without a target, so stage 5 first identifies the financial entity or indicator under discussion. Bangla morphology makes this harder than a string match, because case and definiteness are attached as suffixes. The word for bank appears in the corpus as six distinct surface forms totalling 555 occurrences; exact matching treats them as six unrelated words.

Matching therefore operates on stem prefixes over whitespace tokens, so one stem covers the entire inflectional family. A prefix rule is blunt, so each lexicon entry carries an exclusion list. Without one, the interest rate stem **সুদ** matches the country name **সুদান**, and the stem

for good, ভালো, matches the word for love, ভালোবাসা. Longer stems claim their spans before shorter ones, so a specific target is preferred over a generic word nested inside it.

Target type	Pairs	Examples
MACRO_INDICATOR	1,121	মূল্যস্ফীতি, রিজার্ভ, বিনিয়োগ
REGULATOR	980	বাংলাদেশ ব্যাংক, বাজেট
SECTOR	901	গ্যাস, কৃষি, শিল্প
MARKET	107	শেয়ারবাজার, পুঁজিবাজার
Total	3,109	52 lexicon entries, 48 seen in corpus

Table 4: Target detection output

10. Polarity Assignment

10.1 Target conditioned direction

The central modelling decision is that direction words are not polar on their own. Ordinary sentiment analysis treats an increase word as positive, which in economic discourse is wrong roughly half the time. Each target therefore carries an orientation, and a direction cue is multiplied by it.

Orientation	Targets	Meaning	Examples
+1	14	A rise is good news	reserves, remittance, GDP, exports, investment, employment
-1	9	A rise is bad news	inflation, consumer prices, bad loans, foreign debt, capital flight
0	29	An entity, no direction	banks, the government, the energy sector

Table 5: Target orientation

So রিজার্ভ বেড়েছে, reserves have increased, resolves positive, while মূল্যস্ফীতি বেড়েছে, inflation has increased, resolves negative from the same cue. Because each target is scored

independently, one utterance can carry two different labels at once. A sentence level model cannot represent that distinction.

10.2 Scoring and postprocessing

Supervision comes from a written rule set rather than a black box, so every resulting label can be traced to the tokens that produced it. Each of 104 polarity cue stems carries a weight, a kind (evaluative or direction) and a negation flag.

```
# every cue token within 12 tokens of the target
polarity = cue.polarity
if cue.kind == "direction":
    if orientation == 0: skip
    polarity *= orientation
if cue.negatable and negator within 3 tokens:
    polarity *= -1
decay = 1 / (1 + distance / 4)
score += polarity * cue.weight * decay
```

The score is then thresholded: above +0.55 is positive, below -0.55 is negative, and the band between is non evaluative. Two postprocessing rules proved necessary and were found by reading the output.

- **Negation has scope.** A first version flipped any cue near a negator and scored crisis as positive in সংকট কাটছে না, the crisis is not passing. The particle negates the verb, not the noun, and the sentence is still bad news. Only adjectives and verbs are now marked negatable; polar nouns are not.
- **Bangla builds a negated verb complex.** The particle can sit three tokens from the word it negates, as in স্থিতিশীল করতে পারবেন না, you will not be able to stabilise it. The negation window was widened from one token to three.

Utterances flagged as repetition loops, and those under six words, are excluded from supervision rather than labelled neutral. That removes 90 of 3,109 pairs and leaves 3,019 labelled pairs, of which 284 are negative at 9.4 percent, 2,583 are non evaluative at 85.6 percent, and 152 are positive at 5.0 percent. This imbalance governs the choice of metric in section 12 and the results in section 13.

11. Dataset Construction

The corpus is partitioned by episode, never by row. Two utterances from the same talk show share the panel, the topic, the host's phrasing and the same recognition error pattern. A random row split would place utterances from one recording on both sides of the partition, allowing a model to recognise the episode rather than the sentiment. The reported score would rise and none of that rise would survive contact with a new recording.

Split	Episodes	Pairs	Human	Negative	Non eval	Positive
Train	19	2,026	99	164	1,771	91
Validation	3	320	22	24	278	18
Test	4	673	59	96	534	43

Table 6: Partition by episode

The split routine asserts that no episode identifier appears in more than one partition, so an editing mistake fails loudly rather than silently inflating a score. Test episodes were selected to carry as much of the hand labelled reference set as possible, so the headline figure is measured on human labels from recordings the model has never seen. Validation is used only to select the stopping epoch and is never fitted on.

12. Evaluation Methodology

No reference standard existed, so one was created: a stratified sample of 180 target pairs was read in context and labelled by hand, of which 59 fall inside the held out test episodes. The sample is stratified, so its accuracy is not corpus accuracy; macro F1 is the comparable figure.

The class balance governs metric choice. A model that answers non evaluative for every row scores 85.6 percent accuracy on this corpus while having learned nothing, so accuracy is not used as a headline. The reported metrics are:

- **Macro F1**, the headline. It averages F1 across the three classes with equal weight, so the 152 positive rows count as much as the 2,583 neutral ones. The all neutral model scores 0.31.

- **Per class precision and recall.** Macro F1 conceals which class is failing. Separating it is what revealed that positive precision is the weak point.
- **Cohen kappa**, agreement corrected for chance. Raw agreement of 59.4 percent sounds considerably better than the corrected value of 0.392, which is fair agreement rather than good.
- **Confusion matrix**, the only view that shows the direction of the errors. A single F1 cannot distinguish a model that misses negatives from one that invents positives.

Word error rate, diarization error rate and speaker attribution accuracy are deliberately not reported. Each requires a human corrected reference that this project does not have, and computing them from automatic output would compare the system against itself.

13. Results

System	Macro F1	Notes
Majority class baseline	0.177	Always answers non evaluative
Character n-gram classifier	0.466	Trained on rule labels, 2,026 rows
Rule labeller	0.566	The system that produced the training labels
BanglaBERT pair model	not run	Implemented, not yet trained

Table 7: Macro F1 on the 59 held out human labelled pairs

The trained classifier scores below the rule set that generated its training labels, 0.466 against 0.566. This is the most important result in the report. A student model trained on noisy labels cannot exceed its teacher unless it generalises past the teacher's mistakes, and with 2,026 training rows containing only 91 positive examples it has no basis on which to do so.

Class	Precision	Recall	F1	Support
negative	0.850	0.607	0.708	84
non evaluative	0.517	0.484	0.500	64
positive	0.417	0.781	0.543	32
Macro average	0.594	0.624	0.584	180

Table 8: Rule labeller per class, all 180 pairs. Accuracy 0.594, Cohen kappa 0.392

Negative precision is high at 0.850: when the system reports negative sentiment it is usually correct. Positive precision at 0.417 is where the failure concentrates. The confusion matrix locates it precisely.

Human label	Said negative	Said non eval	Said positive
is negative	51	22	11
is non evaluative	9	31	24
is positive	0	7	25

Table 9: Confusion matrix. 33 pairs read by a human as negative or neutral were reported positive

14. Error Analysis

The natural hypothesis is that recognition noise propagates downstream and degrades the sentiment stage. This was tested by stratifying the 180 reference pairs and does not hold.

Stratum	n	Macro F1
Recognition confidence above median	90	0.550
Recognition confidence below median	90	0.619
No leading truncation	161	0.584
Leading truncation present	19	0.562
Short utterance, under 40 words	37	0.522
Long utterance, 40 words or more	143	0.602

Table 10: Performance by transcript quality stratum

The cleaner half scores no better than the noisier half. At 90 pairs per side the difference sits inside the noise, so the honest reading is that recognition confidence does not separate sentiment accuracy on this sample. Utterance length does correlate, which is expected: a 12 token window around the target finds fewer cues in a short utterance. The bottleneck is the label rules, not the transcripts. Three specific causes account for the errors.

14.1 Prescription read as evaluation

The largest single class, 24 of 73 errors. Bangla economic commentary is dense with prescription. A speaker states what should happen and the rule set counts the favourable word as a judgement about the present state.

অর্থনীতি স্থিতিশীল করতে হবে

The economy must be stabilised. This is a demand, not a compliment. The system reads the word for stable and reports positive.

14.2 Negative meaning with no cue word

22 errors. The cue set holds 104 stems, and plain Bangla frequently conveys bad news without using any of them.

বিনিয়োগ বাইশ পয়েন্ট চার পার্সেন্টে নেমে গেছে

Investment has come down to 22.4 percent. Unambiguously negative, and not one word of it appears in the cue list.

14.3 Negation carried by a suffix

Bangla negates the past tense by attaching নি directly to the verb, so no separate particle exists for the negation window to detect. In বাজেট বাস্তবায়ন হয়নি, the budget was not implemented, the system sees the word for implementation and reports positive.

14.4 Structural cause

Underlying all three is class imbalance. With 85.6 percent of rows non evaluative and only 91 positive examples reaching the training set, there is far too little signal for a 110 million parameter model and thin signal even for logistic regression.

15. Limitations

- No word error rate, diarization error rate or speaker attribution accuracy, as no human reference transcript or speaker annotation exists. The 99.5 percent figure is attribution coverage, not accuracy.
- The 180 reference labels were assigned by a single reader. A second annotator and an inter annotator agreement figure are required before these can properly be called gold labels.
- The BanglaBERT pair model is implemented and its checkpoint loads and tokenises correctly, but it has not been trained. No neural result is claimed.
- The rule labels measure agreement with a rule set, not correctness. Only the 180 human labels measure correctness.
- The reference sample is stratified, so its accuracy figure is not corpus accuracy.
- 26 episodes is a small corpus, and the positive class at 152 rows is its thinnest part.
- The repetition penalty diagnosis in section 7.2 is inferred from the configuration and has not been confirmed by a rerun.

16. Future Work

Ordered by expected gain relative to effort.

1. **Hand label 800 to 1,000 pairs with two annotators.** This removes the ceiling rather than raising it, because training on real labels frees the model from the teacher's bias. Two annotators also yield the agreement figure the project currently lacks. Six members at roughly 150 pairs each is a single working session.
2. **Correct the two recognition settings.** Raise the repetition penalty above 1.0 to suppress decoder loops, and cut chunks with a 2 to 3 second overlap to eliminate the truncated first word in 11.3 percent of utterances. Both are configuration changes that clean the input for every subsequent stage.
3. **Separate prescription from evaluation.** The largest single error class. Bangla marks obligation with a small closed set of endings, chiefly করতে হবে, উচিত, দরকার and প্রয়োজন. Detecting these and suppressing the polarity claim is a short rule.
4. **Handle suffix negation.** Matching a verb final নি catches a class of sign inversions that currently enter the training labels uncorrected.
5. **Train the sentence pair model.** It reads whole phrases rather than character n-grams, so it can learn that a fall in investment is negative without that phrase appearing in a cue list. Worth running once real labels exist.
6. **Extend the corpus.** The pipeline runs end to end on new material without code changes, which makes additional episodes the cheapest route to strengthening the positive class.

17. References

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